



SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !



# College of Health Sciences School of Pharmacy

## Chemistry lab report

NAME

ID

1 werkamaw kassa.....UGR1251/3

1. Melkamedil Yihun.....UGR/7219/13

**Date of submission : feb,8,2022 G.C**

**Submitted to : Doctor Meknen**

1.

Feb8,2022 G.C

## **EXPERIMENT NUMBER : 1, Recrystallization**

**Objective** To purify a contaminated solid compound by recrystallization.

**Introduction** : the purification of solids by crystallization is based upon differences in their solubility in a given solvent.in its simplest form the crystallization process consists of ;

A, dissolving the impure substance in some suitable solvent at or near the boiling point and filter the solution while it is still hot.

B, allowing the hot solution to cool thus causing the dissolved substance to crystallize out and

C, separating the crystals from the solution

### **APPARATUS AND MATERIALS used.**

- Erlenmeyer flask Suction/filter flask
- Stemless funnel
- Measuring cylinder
- Filter paper
- Buchner funnel
- Suction/filter flask
- Glass rod
- Iron stand
- Water aspirator
- Clamp
- Iron ring
- Paper box
- Filter paper
- **Chemicals used.**
- Water
- Benzoic acid
- Ice
- Methyl gas

# Chemistry lab report

---

## PROCEDURE:

- 1) 3g of impure benzoic acid was weighed and placed it in Erlenmeyer flask, then 210ml of water was poured.
- 2) the clamp was fixed on the neck of Erlenmeyer flask and heated it on stove. we made a fluted paper and put it on the gravity filtration set up.
- 2) the solution was boiled and after that we started to filter the it while it was hot by small portion until finished.
- 3) The slution was put under ice after it waited to become cool at room temperature.
- 4) The suction filtration set up was prepared and connected it with water aspirator. the mass of filter paper was measured, which was 0.62g and placed it on Buchner funnel.
- 4) the tap water was turned and filtered the crystals by pouring all of the contents of the Erlenmeyer flask into the Buchner funnel.
- 5) a paper box was prepared and put the benzoic acid that was filtered on Buchner funnel with the filter paper.
- 6) After 2 days, we measured the mass of benzoic acid and the filter paper and it was 3.48g.

## RESULT AND DISCUSSION:

To know the mass of benzoic acid we substract the mass of filter paper from the total mass.

Mass of benzoic acid = total mass of benzoic acid + filter paper-mass of filter paper

Mass of benzoic acid = 3.48g – 0.62g

Mass of benzoic acid= 2.86g.

- ✓ The percentage impurity of our sample can't be calculated because of the error of balance.

## QUIZ QUESTIONS AND THEIR ANSWERS;

1)How does a water aspirator create partial vacuum?

- ✓ When water from the tap is forced at a high velocity through a small nozzle, at the smallest side of the nozzle a vacuum is created.

# Chemistry lab report

---

2) what is the significance of the term "like dissolves like"?

- ✓ The term is useful for us to choose the right solvent.

3) what are the difference between an ordinary paper and filter paper?

- ✓ The filter paper is relatively strong and it has a tiny pores that allows fluid pass through it. Since it is made of fibers.

5) some solids make perfect crystals, while others are amorphous. what may be the cause for this?

- ✓ The cause for some substances to be amorphous and others to be crystalline is the atomic arrangement difference. as for crystalline they are arranged in definite and repeating 3D patterns in highly ordered microscopic structure.

## **EXPERIMENT; 2; Simple distillation**

**Objective** to purify contaminated liquid sample by simple distillation.

**Introduction;** Distillation is one of the most common methods of purifying liquids. if a liquid sample is contaminated with impurities that are non volatile, or if it is contaminated with another liquid which boils at least 50°C higher than the boiling point of the desired liquid, the main liquid can be obtained in a pure form by simple distillation. thus, simple distillation is useful when only a small amount of impurity is present or when the liquid to be separated have widely differing boiling points.

Simple distillation involves refining a liquid through the process of separation. the most common purpose of simple distillation is to purify drinking water of unwanted chemicals and minerals such as salt. most commonly it is used in the production of alcoholic beverages such as vodka and whiskey.

### **APPARATUS AND MATERIALS;**

Distillation flask  
Cork  
Condenser  
Bunsen burner  
Iron stand  
Thermometer  
Measuring cylinders

# Chemistry lab report

---

Boiling chips  
Iron bowl  
Iron ring  
Clamp  
Bosshead  
Thin plastic tube

## **Chemicals used**

Water  
Ethanol

## **PROCEDURE:**

- 1). 2 iron stands were put near to each other. the one was fixed with iron ring by bosshead. Iron bowl was put on the ring. Above the iron ring clamp fixed by using bosshead putting the distillation flask partially in to the iron bowl.
- 2). Condenser was attached with the distillation flask by plastic cork. other end of condenser was attached with another iron stand by clamp which was attached by bosshead to the stand.
- 3). 2 thin plastic tubes were attached with the 2 outlets of the condenser. The other end of upper attached tube fixed with water pipe. and the other tube end left as an outlet.
- 4) 25 ml of contaminated ethanol was poured into the distillation flask with 3 boiling chips and then covered with a cork having thermometer at the middle.
- 5). Water was poured into iron bowl and measuring cylinder was placed at the bottom of condenser lower end.
- 6). We ignited the Bunsen burner and placed it under the iron bowl.
- 7). Then the water started boiling, first condensed drop of ethanol was observed at 70°C. the process proceeds until the last drop of ethanol was dipped at 81°C.  
Finally 20ml of pure ethanol was found in the measuring cylinder.

**This table shows the observed difference in boiling point at each 2ml interval.**

## Chemistry lab report

|             |               |
|-------------|---------------|
| <u>2ml</u>  | <u>70°C</u>   |
| <u>4ml</u>  | <u>70°C</u>   |
| <u>6ml</u>  | <u>70°C</u>   |
| <u>8ml</u>  | <u>70°C</u>   |
| <u>10ml</u> | <u>71°C</u>   |
| <u>12ml</u> | <u>71.4°C</u> |
| <u>14ml</u> | <u>72°C</u>   |
| <u>16ml</u> | <u>73°C</u>   |
| <u>18ml</u> | <u>73°C</u>   |
| <u>20ml</u> | <u>81°C</u>   |

### RESULT AND DISCUSSION;

25 ml of impure ethanol was added into the distillation flask. after distillation 20ml of pure ethanol was found. this shows that 5ml was the impurity. Ethanol and the impurity have highly differ in boiling points minimum up to 50°C. it was observed that the boiling point of ethanol was reduced by the impurity from 81°C to 70°C. and final temperature was 81°C.

The impurities were left in the distillation flask having a deep red colour.

### QUIZ QUESTIONS AND ANSWERS;

1) Why doesn't all the liquid vaporize at once when the boiling point is reached?

Because of boiling point difference in our case the impurities have higher boiling point than that of the boiling point reached.

2) An organic liquid begins to decompose at 80°C . it's vapor pressure at that temperature is 36mm.how would you distill the liquid?

We use a zeotropic distillation in which we add toluene(boiling at 110.6°C) to the substance.

3) Why is cold water circulated through a condenser from the bottom upward rather than from the top to down ward?

The water is circulated from bottom to top to have even cooling .the condenser at the upper part will be hotter than the bottom as we go down it will be colder and colder. when the colder water moves up it will be hotter than the bottom. therefore collision of hot and cold will be avoided.so, the upper part will be cooled as the cold water passes continuously.this also helps to condense the vaporized liquid.

4) If the liquid to be distilled had a boiling point higher than 90°C. what kind of bath will you use?

We would use alternative baths such as oil bath, silicon bath or sand bath.

5) Why is it that when a pure liquid is boiled its temperature remains constant?

The temperature of pure substance remains constant during its melting and boiling point. because the change in any state of matter as solid to liquid or the liquid to a gas involves crossing the latent heat of fusion, which causes a change in intermolecular spacing of the molecule.

## **EXPERIMENT ;3 Survey of some functional groups**

**Introduction;** a functional group is a unique combination of atoms present in a molecule. That is particularly responsible for the characteristic set of reactions that the molecule exhibits. Some of the most important functional groups include hydroxyl, olefin, ketone, aldehyde, amine, carboxylic acid, ester, acid halide, anhydride, nitrile, ether, halide, aromatic ring etc. It is possible to distinguish by means of simple tests one functional group from the other.

Alkanes, the simplest of all organic compounds lack the typical functional groups making them most unreactive. They can be inert even in the presence of concentrated sulphuric acid. Alkanes react slowly or not at all with bromine in the dark. However, in the presence of light they react fairly rapidly. This can be explained by means of radical chemistry.

The simplest of alkanes is methane, which is a gas at room temperature, so are ethane, propane, and butane. Pentane and hexane are low boiling liquids and are among the most useful solvents in laboratory as well as in industry. Mixtures of alkanes obtained during fractional distillation of petroleum are called petroleum ether or simply petrol. They are usually categorized by means of their boiling point range. A low boiling petroleum ether has boiling point of 40-60°C, while the higher boiling ones are 60-80°C and 80-120°C.

Higher boiling Naphta or kerosene is the most common sources of energy. This fuel is mixture of alkanes such as isomeric pentanes, hexanes, heptanes, and octanes and small amounts of unsaturated and aromatic hydrocarbons.

### **APPARATUS AND MATERIALS;**

- Small test tubes
- Litmus paper
- Droppers
- N-hexane
- Sulfuric acid
- Water
- Ethanol
- Petroleum ether
- Solution of bromine in carbon tetra chloride or bromine water
- Cyclohexene
- Potassium permanganate
- Benzene
- Toluene
- Ice
- Nitric acid

# Chemistry lab report

---

DNPH

Acetone

Phenol

Ferric chloride solution

Kerosene

Calcium carbide

PROCEDURE;

## 1. ALKANES

To test solubility of alkanes 4 small test tubes were used and N-Hexane was poured into all test tubes. Then we added 3ml of water, ethanol, petroleum ether and sulfuric acid in each test tubes respectively.

In separate test tubes 1ml portion of petroleum ether, N-Hexane and kerosene were poured. 4-5 drops of solution of bromine in carbon tetra chloride was added into each. then place in dark for 15 minutes.

In separate test tubes 1ml portion of petroleum ether, N-Hexane and kerosene were poured. 4-5 drops of solution of bromine in carbon tetra chloride was added into each. then place in light for 5 minutes. blue litmus paper was fixed on the mouth of each test tubes to test solubility reaction in light.

In separate test tubes 1ml portion of petroleum ether, N-Hexane and kerosene were poured. then 1ml of potassium permanganate was added in each test tube to test solubility.

## ALKENES

0.5 ml of cyclohexene was poured into a test tube. then 3 drops of sulfuric acid was added into 1 test tube to test solubility.

0.5 ml of cyclohexene was poured into a test tube. then 3 drops of bromine water was added to test if they react.

0.5 ml of cyclohexene was poured into a test tube. then 1 ml of potassium permanganate was added into it to test reaction.

## ALKYNES

Acetylene generator was set up by our instructor in the fume hood. then water drop was allowed drop by drop into a calcium carbide to test if acetylene was generated or not. bromination test (with bromine water) and baeyer's test (with potassium permanganate).

## AROMATIC HYDROCARBONS

6 drops of benzene were added in the test tube then 3 drops of bromine water was added to test the reaction of unsaturated.

In test tube nitrating mixture by adding 10 drops of concentrated sulfuric acid, 3 drops of concentrated nitric acid and 4 drops of toluene together. then we shook the mixture and put it for 15 minutes.



# Chemistry lab report

---

## KETONE

10 drops of 2,4-dinitrophenyl-hydrazine(DNPH) was placed in a test tube. then, 10 drops of acetone poured to observe if precipitate was formed.

## PHENOL

20 drops of phenol was poured to a test tube and 3 drops of neutral ferric chloride solution was added into that test tube.

### RESULT AND DISSCUSSION:

Note that

“n” represents no reaction for reaction test and insoluble for solubility test.

“y” represents there exist a reaction for reaction test and soluble for solubility test.

#### 1)ALKANE

| Alkane   | water | ethanol | Petroleum ether | Sulfuric acid |
|----------|-------|---------|-----------------|---------------|
| N-Hexane | n     | y       | y               | n             |
| kerosene | N     | y       | y               | n             |

| Alkane          | Dark reaction                                          | Light reaction                                         | Effect of $\text{KMnO}_4$ |
|-----------------|--------------------------------------------------------|--------------------------------------------------------|---------------------------|
| Petroleum ether | n and no change of litmus paper also color of solution | Y and blue litmus turned red also solution decolorized | N                         |
| N-Hexane        | n and no change of litmus paper also color of solution | n and no change of litmus ,solution slightly faded     | N                         |

## II.

### 2)ALKENE

# Chemistry lab report

|          |                                                        |                                                        |   |
|----------|--------------------------------------------------------|--------------------------------------------------------|---|
| Kerosene | n and no change of litmus paper also color of solution | Y and blue litmus turned red also solution decolorized | N |
|----------|--------------------------------------------------------|--------------------------------------------------------|---|

## 2)ALKENE

|             |               |               |                   |
|-------------|---------------|---------------|-------------------|
| Alkene      | Sulfuric acid | Bromine water | KMnO <sub>4</sub> |
| cyclohexene | y             | y             | y                 |

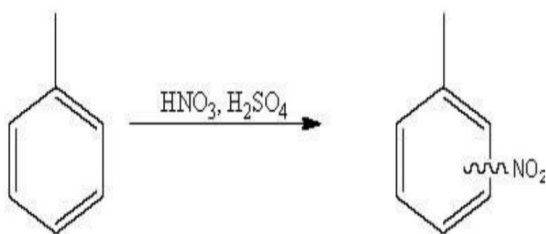
## 3)ALKYNE

When calcium carbide reacted with water, acetylene was generated and was proved that it existed because when the gas outlet put in bromine water and potassium permanganate decolorized both. this indicates reaction took place between them. and also the presence of acetylene was seen when match lighted near it created acetylene welding flame.

## 4)AROMATIC HYDROCARBONS

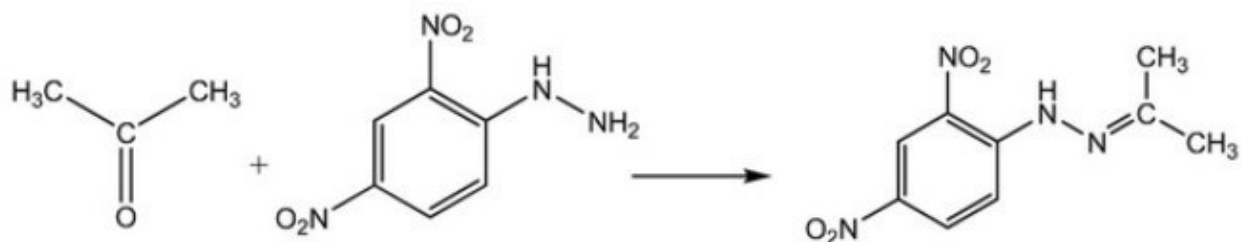
a) When benzene and bromine water were poured together, we have seen no color change. that indicate there was no reaction.

b) the solution formed from sulfuric acid, nitric acid and toluene was poured into a large test tube containing piece of ice. nitro benzene is formed as slight yellow oil.



## 5)KETONE

When DNPH and acetone were reacted, they formed noticeable precipitate.



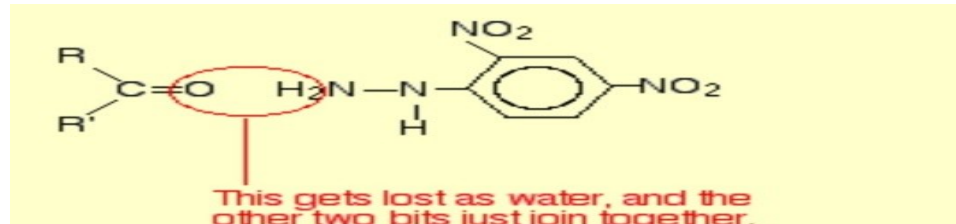
### 6)PHENOL

When colorless phenol added to a color less neutral ferric chloride solution,resulted a purple final substance.

#### QUIZ QUESTIONS AND THEIR ANSWERS

Give the mechanism of the reaction of DNPH with acetone?

In terms of mechanisms, this is a nucleophilic addition elimination reaction.the DNBH first adds across the carbon-oxygen double bond(the addion stage) to give an intermediate compound which then loses a molecule of water(the elimination stage).



What is the difference between addition and substitution reaction?

Addition reaction occurs when 2 or more reactants combine to form a single products.this product will contain all the atoms that were present in a reactants.whereas the substitution reaction occurs when an exchange of elements in a reactant take place.atom or ion in the molecule is replaced by another atom, ion or group.

Give mechanism for the nitration of toluene?

Electrophilic aromatic substitution represents an important class of reactions in organic synthesis. In "aromatic nitration" aromatic organic compounds are nitrated via an electrophilic aromatic substitution mechanism involving the attack of the electro-rich benzene ring on the nitronium ion.the formation is shown below.

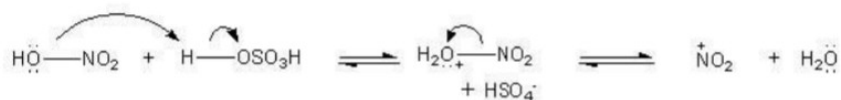
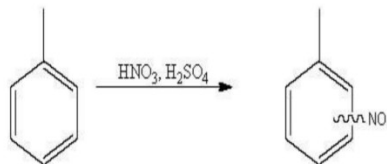


Figure 1: The mechanism for the formation of a nitronium ion.

**Toluene undergoes nitration to give ortho and para nitrotoluene isomers, but if it is heated it can give dinitrotoluene and ultimately the explosive trinitrotoluene(TNT).**



### Experiment 4 *Preparation of aspirin (Acetylsalicylic acid)*

**Objective** to prepare acetyl salicylic acid, commonly known as “Aspirin”

**Introduction.** Aspirin also known as acetylsalicylic acid (ASA): is a medication used to reduce pain, fever, or inflammation. To prepare aspirin, salicylic acid is reacted with excess of acetic anhydride. Small amount of a strong mineral acid or a weak base can be used as a catalyst. To speed up the reaction it is necessary to warm the reaction mixture using gentle heat such as placing the mixture over a water bath.

# Chemistry lab report

---

## Procedure

- 1). 2gm of salicylic acid was measured and prepared by our instructor; this amount was put into a dry test tube.
- 2). 2ml of acetic anhydride was added to this test tube, and then following that we added 5 drops of conc. Sulphuric acid.
- 3). Water was added to a water bath, and heated by a Bunsen burner.
- 4). When it started to warm, the test tube containing the salicylic acid, acetic anhydride and sulphuric acid was partially immersed into the hot water bath. Using a stirrer the mixture was stirred.
- 5). 60ml of water was measured using a measuring cylinder, and added to a beaker.
- 6). The mixture was added to the beaker containing the 60ml.
- 7). The beaker was placed into an ice bath for the process of crystal formation. After a few minutes crystals started forming slowly.
- 8). A suction filter was assembled, and a filter paper was put on top. The mixture that formed crystals was transferred into the suction filter.
- 9). The product was left on the filter paper. It was dried using paper for the next reactions of solubility.

### Solubility test of aspirin

After drying the product we filtered in the previous reaction, the resulting product was collected. 3 dry test tubes were prepared. 1ml of Water, ethanol, and sodium bicarbonate was added into the three different test tubes. A small amount of the product aspirin was put on to the tip of a spatula and it was added to the three different test tubes. The results of each was observed and recorded.

### Solubility test of salicylic acid

3 dry test tubes were prepared. 1ml of Water, ethanol, and sodium bicarbonate was added into the three different test tubes. A small amount of salicylic acid was put on to the tip of a spatula and it was added to the three different test tubes. The results of each was observed and recorded.

### Test for phenolic hydroxyl group

## Chemistry lab report

---

1 drop of 1% ferric chloride solution was added to the above test tubes that were used to test for solubility. The result was observed and recorded.

### Apparatus used

Water bath  
Bunsen burner  
Clamp  
Boss head  
Suction filter  
Filter paper  
Test tube  
Stand  
Beaker  
Test tube holder  
Spatula  
Stirrer  
Measuring cylinder

### Chemical used

Water  
Ethanol  
Sodium bicarbonate  
Conc. Sulphuric acid  
Acetic anhydride  
Aspirin

### Result and discussion

|                | solvents    |         |                    |
|----------------|-------------|---------|--------------------|
| sample         | water       | ethanol | NaHCO <sub>3</sub> |
| Aspirin        | Not soluble | soluble | Soluble            |
| Salicylic acid | Not soluble | soluble | soluble            |

The experiment result shows that the solubility of Aspirin( Acetyl salicylic acid) in ethanol and NaHCO<sub>3</sub> is high when compared to water. Aspirin has a limited solubility in water, the hydrophobic nature of aspirin makes it insoluble.

Salicylic acid is soluble in water, and it releases carbon dioxide.

## Chemistry lab report

|                | Solvent                             |                                           |                                          |
|----------------|-------------------------------------|-------------------------------------------|------------------------------------------|
| Sample         | water                               | ethanol                                   | NaHCO <sub>3</sub>                       |
| Aspirin        | Reaction( from colorless to purple) | Reaction( from colorless to light purple) | Reaction( from colorless to yellow)      |
| Salicylic acid | Reaction( from colorless to purple) | Reaction( from colorless to purple)       | Reaction( from colorless to pink/orange) |

When ferric chloride solution is added to a test tube that contains Aspirin and water, it changes the color from colorless to purple, when it is added to Aspirin and NaHCO<sub>3</sub> solution it changes its color from colorless to yellow. **However when we add ferric chloride in Aspirin and ethanol, it changes its color from colorless to light purple, but this result should have been colorless. The result recorded on the table occurred due to the contact between the dropper that was previously used for salicylic acid.**

When ferric chloride solution is added to a test tube containing salicylic acid and ethanol also salicylic acid and water, both change their color from colorless to purple. However when we add it to a solution of salicylic acid and NaHCO<sub>3</sub> it changes its color from colorless to pink/orange.

### Question and answer

1). Why is the carboxyl group of salicylic acid not acetylated?

During the synthesis of aspirin, the carboxylic group of salicylic acid is not acetylated; this is because of the lower nucleophilicity of the carboxylic group.

That's why acetic anhydride prefers phenolic-OH group rather than carboxylic group. Carboxylic acids are not as nucleophilic as phenols. It is possible to form a mixed anhydride of a carboxylic acid, but likely the conditions used were either not strong enough or it was formed and was hydrolyzed upon workup ( mixed anhydrides are also not too stable).

2). What is the role of the acid catalyst? Also a base like pyridine can be used for the catalysis. Explain the role of the catalyst by giving the mechanism of the reaction.

Just a few drops of sulphuric acid added to the mixture means that there are some free positive hydrogen ions in the solution which can bind to the acetic anhydride. The acid serves to protonate one double bonded oxygen on the acetic anhydride this makes the reaction with the salicylic acid faster. The proton is returned to the acid in the end so it is just a catalyst.

## Chemistry lab report

---

Pyridine is more nucleophilic than the alcohol, and it attacks the acyl chloride rapidly, forming a highly electrophilic (because of the positive charge) intermediate. Because pyridine is acting as a nucleophile to speed up the reaction, yet is unchanged by the reaction, it is a "nucleophilic" catalyst.

The melting point of salicylic acid is 157°C-159°C while that of aspirin, which has a higher molecular formula, is around 135°C. Explain why?

The Carboxylic acid group in aspirin's structure allows molecules of aspirin to form hydrogen bonds with each other. Because of the permanent dipole bonds within the aspirin molecule. Because of such a strong attraction, it indicates that aspirin's melting point is relatively high. **However, the melting point of salicylic acid is expected to be higher than the melting point of aspirin as there are more hydrogen bonds formed which means that more energy is needed to break the bonds. So when testing the purification of aspirin, pure salicylic acid's melting point should be higher than the aspirin sample.**

3). What is the rationale for the observed color reaction between salicylic acid and ferric chloride?

$\text{FeCl}_3$  (aq). Iron (III) ion reacts with phenols to form a purple complex. Salicylic acid contains a phenol group, but acetylsalicylic acid does not. Therefore, if you add  $\text{FeCl}_3$  to an aspirin sample and you see a purple color, it means that there is still some salicylic acid present and the sample is impure.  $\text{FeCl}_3$  will not react with aspirin. It will, however, react with salicylic acid, which is used to synthesize aspirin.  $\text{FeCl}_3$  (aq). Iron (III) ion reacts with phenols to form a purple complex. Salicylic acid contains a phenol group, but acetylsalicylic acid does not. A purple colour is an indication of salicylic acid. A sample of pure aspirin should not exhibit a colour change.



## Chemistry lab report

---

DARE;Jan,25,2021G.C

## Chemistry lab report

---

Date;feb,01,2021 G.C